ME-410: Mechanical Product Design and Development

Reconfigurable Robotics Lab

HAPTIC GUIDANCE SYSTEM

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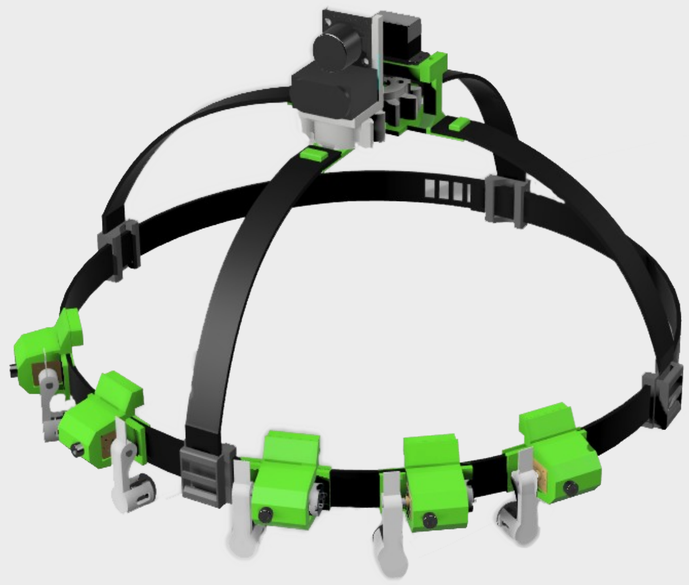
Motivation

⇒ Increase safety & independence with a wearable device

Mobility is a fundamental human right, but current aids (such as the white cane) face critical limitations:

- **Overhead Safety:** Fail to detect high/distant obstacles → high injury risk (trips, collisions).
- **Hands-Free:** Requires manual/auditory input → increased cognitive load & distraction.

→ **Our Goal:** Provide rich, intuitive spatial awareness via a complementary, hands-free, wearable system. We translate complex data into simple, actionable haptic feedback on the forehead to close the safety gap and maximize user independence.



Omni Sense

Omni-directional sensing for visually impaired individuals

Design

⇒ Integrates multiple sensor modalities into a compact, wearable headband platform designed for comfort and robust environmental perception.

The **core sensing and detection unit**:

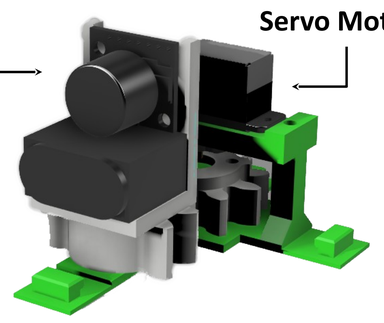
- 1 Laser Time-of-Flight (ToF) sensor deliver precise distance measurements and wide field of view
- 1 ultrasonic ToF sensors complements laser ToFs by extending range and improving reliability of different surface detections

Ultrasonic ToF

Laser ToF

are mounted on a motorized rotating platform creates a comprehensive 160-degree field of view combining.

Servo Motor



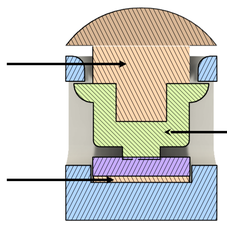
The **haptic feedback unit**:

- Composed of 5 DC motors with articulated arms and pressure pads to deliver intuitive haptic feedback.
- Pressure intensity correlates with obstacle proximity, offering a natural sense of direction and distance.

Lever pre-load design

Contact skin part

Pressure pad (orange)



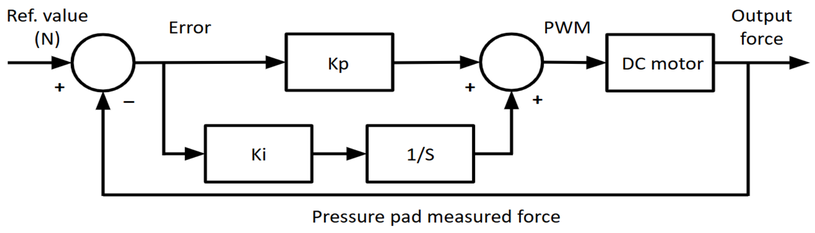
Pin

ensures a good concentration of the force

The **wearable hardware**:

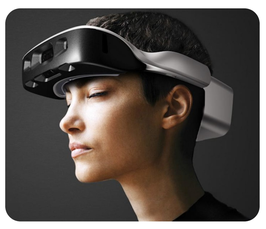
- The sensor assembly is housed in a lightweight 3D-printed enclosure with custom mounting brackets.
- A flexible TPU headband ensures comfortable fit for extended wear, with 3D-printed clips securing the subsystem and a glide buckle enabling quick adjustments.

The **closed loop** used for control :



⇒ PI controller using pressure pad feedback as the only sensing input to create a consistent tactile pressure on the user's skin, independent of wearing tightness or battery level.

State of the art



The field of electronic travel aids (ETAs) is evolving, featuring devices like the Smart Cane or Wearables (Biped AI), primarily use audio or vibration. Vibration offers poor resolution, often relying on complex frequency patterns to convey distance, and is often placed on areas with low tactile sensitivity (chest/abdomen).

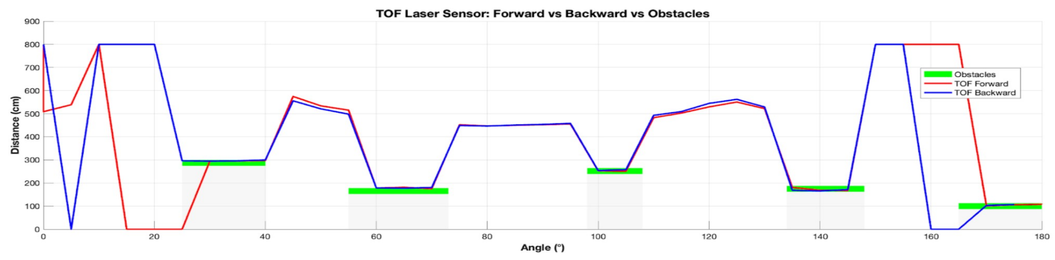
The Gap: Current systems fail to provide intuitive, scalar haptic feedback that cleanly separates distance and direction, leading to high cognitive load and slow reaction times.

Our Innovation: We introduce a mechanically-scaled haptic system on the high-acuity forehead area. This uniquely decouples the information: 5 motor positions signal Direction, while controlled Pressure provides a continuous, intuitive measure of Proximity.

Performance

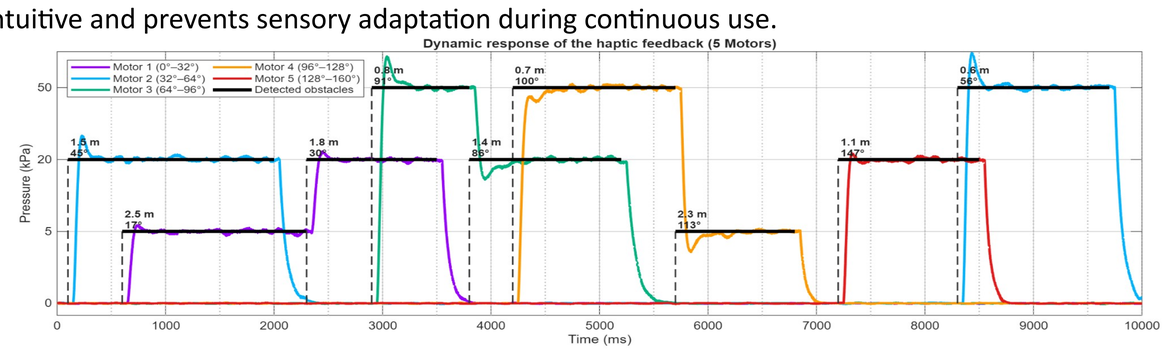
Environmental Perception Accuracy

This plot validates the **precision of the rotating Laser ToF sensor** assembly. The data demonstrates a strong linear correlation between the measured sensor output and the actual obstacle placement across the entire 160° field of view. The sharp signal edges confirm the system's ability to distinctly resolve adjacent obstacles and filter noise, ensuring a reliable real-time spatial mapping for the user.



Dynamic Haptic Response

This temporal profile illustrates the **closed-loop pressure control algorithm**. Distance data is dynamically translated into three distinct pressure tiers (approx. 5, 20, and 50 kPa) to encode proximity. The specific tuning includes calibrated damping and transient behaviors, such as the undershoot for receding objects, to create "organic" tactile cues. This ensures the feedback is intuitive and prevents sensory adaptation during continuous use.



Summary and Future Development

⇒ The **OmniSense** headband successfully demonstrates a scalable, and intuitive solution to the ETAs gap

ACHIEVED	TO COME
✓ Provide immediate **Direction** (via 5 motor positions) and **Proximity information** (via motor pressure)	★ Wider range of detection through LiDar implementation
✓ Compact, wearable device	★ Adding Object Identification to filter alerts
✓ Mechanically-scaled haptic pressure	★ GPS/IMU to fuse pathfinding with local obstacle avoidance
✓ Validation by the closed-control-loop's stable performance	★ Design : robust miniaturization and ergonomic optimization

EPFL

References:

1. C. Panazan and E. Duft, "Intelligent Cane for assisting the visually impaired", May 2024
Biped ai , "NOA by biped ai - AI mobility ves for he blind". Available: https://biped.ai/eu/
3. J. Lundälv and J. Theodelius, "Risk of Injury Events with patients with visual imparements", September 2021